

T1: BASIC IMAGE OPERATIONS

1. Cross-correlation H F and convolution H*F differ: convolution additionally:
- A. Uses a larger kernel
 - B. Flips H both horizontally and vertically before computing the inner product
 - C. Requires weights summing to 1
 - D. Only works on grayscale
2. A 2D Gaussian $G(x,y;\sigma)=G(x;\sigma)\cdot G(y;\sigma)$ is separable. The key computational saving:
- A. Reduces filter size from k^2 to k
 - B. Allows a single 2D pass
 - C. Costs $O(2k)$ multiplications per pixel instead of $O(k^2)$
 - D. Enables integer arithmetic
3. F convolved with centered impulse δ : result $G=F*\delta$ equals:
- A. ∇F
 - B. All-zero image
 - C. F itself (identity operation)
 - D. Blurred F
4. Gradient direction $\theta=(F_y/F_x)$ at a pixel is:
- A. Parallel to the nearest edge
 - B. Perpendicular to the nearest edge (the edge normal direction)
 - C. Always toward brighter regions
 - D. Equal to the second-derivative magnitude
5. The gradient magnitude $|\nabla F|$ is computed as:
- A. $|F_x|+|F_y|$
 - B. $(|F_x|,|F_y|)$
 - C. $\sqrt{(F_x^2+F_y^2)}$
 - D. $F_x\cdot F_y$
6. Increasing σ in $G(x,y;\sigma)=(1)/(2\pi\sigma^2)e^{-((x^2+y^2)/(2\sigma^2))}$:
- A. Sharpens edges
 - B. Reduces the kernel's spatial extent
 - C. Removes more high-frequency content and blurs over a larger region
 - D. Has no effect on a noise-free image
7. A filter with weights summing to zero, e.g. $[-1,0,1]$, is a:
- A. Low-pass filter preserving mean intensity
 - B. High-pass / difference filter removing the DC component
 - C. Identity filter
 - D. Gaussian smoothing filter
8. Central finite difference for $\partial F/\partial x$ at pixel (i,j) :
- A. $F[i+1,j]-F[i,j]$
 - B. $(F[i+1,j]-F[i-1,j])/2$
 - C. $F[i,j]-F[i-1,j]$
 - D. $(F[i+1,j]+F[i-1,j])/2$
9. Convolving $G(\sigma_1)$ with $G(\sigma_2)$ yields $G(\sigma)$ where $\sigma=$:
- A. $\sigma_1+\sigma_2$
 - B. $\sigma_1\cdot\sigma_2$
 - C. $\sqrt{(\sigma_1^2+\sigma_2^2)}$
 - D. $|\sigma_1-\sigma_2|$
10. Why smooth BEFORE computing image gradients?
- A. Smoothing increases gradient magnitude at true edges
 - B. Differentiation multiplies by $i\omega$ in Fourier domain — amplifying high- ω noise; smoothing suppresses it first
 - C. Smoothing makes gradient direction nonlinear
 - D. Order doesn't matter
11. Associative property $F*(G*H)=(F*G)*H$ is useful because:
- A. Enables parallel pixel processing
 - B. Sequential filters can be pre-combined into one kernel — one image pass instead of multiple
 - C. Proves convolution equals cross-correlation
 - D. Eliminates boundary effects
12. Box filter vs. Gaussian of similar support for noise removal: box is worse because:
- A. Requires more computation
 - B. Its DFT is a sinc with sidelobes — high frequencies not fully attenuated
 - C. Cannot handle color images
 - D. Amplifies shot noise
13. Quantization (rounding to nearest integer) introduces per-pixel error of at most:
- A. $\Delta/2$ (half spatial sampling step)
 - B. ± 0.5 gray levels
 - C. Proportional to total pixel count
 - D. Zero for large dynamic range
14. A color image is a function $f:\mathbb{R}^2\rightarrow?$
- A. $\mathbb{R}$
 - B. $\mathbb{R}^2$
 - C. $\mathbb{R}^3$
 - D. $[0,255]^3$ exclusively
15. Zero-padding at image borders introduces:
- A. Random noise near borders
 - B. Darkening near borders — filter averages genuine pixels with exterior zeros
 - C. Edge sharpening near borders
 - D. Repeating tile pattern
16. Convolution is commutative ($H*F=F*H$) but cross-correlation is not. Why?
- A. Convolution uses larger kernels
 - B. The kernel flip in convolution makes the operation symmetric under exchange of arguments
 - C. Cross-correlation produces larger outputs
 - D. Convolution defined only for square kernels
17. The 'derivative of Gaussian' filter $(d)/(dx)G_\sigma$ is obtained by differentiating the Gaussian. Compared to computing $(d)/(dx)(G_\sigma^*I)$ separately, using a single convolution with the pre-computed $(d)/(dx)G_\sigma$ kernel is justified by:
- A. The Gaussian derivative is always separable into 1D filters
 - B. Associativity of convolution: $(d)/(dx)(G_\sigma^*I)=(d)/(dx)(G_\sigma^*)I$

- C. The derivative of a Gaussian has fewer non-zero coefficients than G_σ
 - D. Differentiation is a nonlinear operation and must be applied last
18. A cascade of two linear filters H_1 then H_2 equals:
- A. H_1*F+H_2*F
 - B. $(H_1*H_2)*F$ — one convolution with the combined kernel
 - C. $(H_1+H_2)*F$
 - D. Undefined without specifying order
19. A filter with weights summing to 1 (e.g. normalized Gaussian):
- A. Removes all DC content
 - B. Preserves mean intensity — DC gain =1
 - C. Acts as an edge detector
 - D. Amplifies high frequencies
20. Padding strategy comparison: zero-padding vs mirror (reflect) padding at image borders. Which statement is correct?
- A. Zero-padding always gives higher-accuracy gradients at borders
 - B. Mirror padding replicates edge pixels symmetrically — avoids the artificial dark halo introduced by zero-padding
 - C. Both give identical results for symmetric kernels like the Gaussian
 - D. Mirror padding is only valid for filters with odd kernel size

T2: EDGE DETECTION

1. Zero-crossings of $\nabla^2(G*F)$ locate edges because:
- A. They mark where the first derivative is maximum
 - B. They mark where $\nabla^2(G*F)$ changes sign — the inflection point of the smoothed intensity profile
 - C. They mark where gradient magnitude is zero
 - D. They mark local intensity minima
2. The Sobel x-kernel $[-1\ 0\ 1\ -2\ 0\ 2\ -1\ 0\ 1]$ is the outer product of:
- A. $[-1,0,1][1,1,1]^T$
 - B. $[-1,0,1][1,2,1]^T$
 - C. $[1,2,1][-1,0,1]^T$
 - D. $[-1,1][1,2,1]^T$
3. Canny's non-maximum suppression (NMS) step:
- A. Applies Gaussian to remove noise
 - B. Thins edges to 1 pixel: zeros pixels that are not local maxima along the gradient direction
 - C. Removes weak edges via a single threshold
 - D. Converts gradient magnitude to binary
4. Hysteresis uses T_{high} and T_{low} . A weak pixel ($T_{low}\leq|\nabla|<T_{high}$) is kept only if:
- A. Connected (8-neighbor) to a strong pixel ($|\nabla|\geq T_{high}$)
 - B. Its gradient direction is H or V
 - C. Local mean gradient exceeds T_{low}
 - D. It appears in 3+ consecutive frames
5. Efficiency of the derivative of Gaussian $(d)/(dx)(G*F)=(d)/(dx)G*F$:
- A. $(d)/(dx)G$ is a smaller kernel
 - B. By associativity: smooth-then-differentiate = apply the pre-computed derivative kernel in one pass
 - C. Avoids zero-crossings
 - D. Equals the Sobel filter exactly
6. The 2D Laplacian operator:
- A. $\nabla^2 f=(\partial f)/(\partial x)+(\partial f)/(\partial y)$
 - B. $\nabla^2 f=(\partial^2 f)/(\partial x^2)+(\partial^2 f)/(\partial y^2)$
 - C. $\nabla^2 f=((\partial f)/(\partial x))^2+((\partial f)/(\partial y))^2$
 - D. $\nabla^2 f=(\partial^2 f)/(\partial x\partial y)$
7. Increasing σ in the Canny detector:
- A. Detects more fine-scale edges including noise
 - B. Detects fewer, more reliable edges at coarser scale — fine texture disappears
 - C. Requires additional hysteresis thresholds
 - D. Has no effect on detected edges
8. The Laplacian of a uniform intensity region gives:
- A. A large positive value
 - B. Zero — all second derivatives vanish
 - C. The mean intensity value
 - D. An unpredictable nonzero result
9. The slides show two complementary edge detection approaches: the Laplacian of Gaussian (LoG) and the derivative of Gaussian. What is the key operational difference?
- A. LoG detects edges via zero-crossings of $\nabla^2(G*F)$; derivative of Gaussian detects edges via peaks in $(d)/(dx)(G*F)$
 - B. LoG works only on color images; derivative of Gaussian works on grayscale
 - C. LoG is a first-derivative method; derivative of Gaussian is a second-derivative method
 - D. LoG requires a larger σ than the derivative of Gaussian
10. Differentiating an image amplifies noise because:
- A. Noise has lower frequency than signal
 - B. Differentiation multiplies by $i\omega$ in Fourier domain — amplifying high- ω components where noise dominates
 - C. Differentiation adds quantization error
 - D. Pre-smoothing changes pixel values
11. Canny NMS uses $\theta\ 180^\circ$ quantized to 4 directions because:
- A. Edges are undirected — ∇I and $-\nabla I$ both describe the same edge boundary
 - B. Always choose the direction with larger gradient component
 - C. Sign of F_y resolves the ambiguity
 - D. NMS run twice with θ and $\theta+180^\circ$
12. A zero-sum Laplacian kernel (center +4, neighbors -1) ensures:
- A. Output image has same mean as input
 - B. Zero response in constant regions — correct edge detection behavior
 - C. The kernel is separable
 - D. Integer overflow is prevented
13. The slides describe the Sobel kernel as a 'common approximation of the derivative of Gaussian.' The Sobel x-direction kernel is:

- A. $[-1\ 0\ 1\ -1\ 0\ 1\ -1\ 0\ 1]$ — uniform vertical weighting
 - B. $[-1\ 0\ 1\ -2\ 0\ 2\ -1\ 0\ 1]$ — center row weighted more heavily
 - C. $[1\ 2\ 1\ 0\ 0\ 0\ -1\ -2\ -1]$ — detecting vertical change
 - D. $[-1\ -2\ -1\ 0\ 0\ 0\ 1\ 2\ 1]$ — detecting upward change
14. At a pixel where $\nabla^2(G*F)>0$, the intensity profile in the smoothed image is:
- A. At a local maximum (bright spot)
 - B. At a local minimum (dark relative to surroundings) — concave upward
 - C. On a steep edge boundary
 - D. In a perfectly uniform region
15. In Canny edge detection, why might you apply the detector at multiple values of σ ?
- A. Larger σ removes more noise but also blurs fine details — different σ find edges at different spatial scales
 - B. Using multiple σ is required for hysteresis thresholding to work
 - C. Multiple σ are necessary to compute the gradient direction accurately
 - D. The Canny algorithm is only valid when $\sigma=1$; other values require a different algorithm
16. Simple thresholding of $|\nabla F|$ gives thick edges. Canny avoids this by:
- A. Using a larger σ
 - B. NMS — keeping only local maxima along gradient direction, yielding single-pixel edges
 - C. Morphological thinning post-hoc
 - D. Two-sided threshold
17. A horizontal edge (bright above, dark below): which gradient component is large?
- A. $\partial F/\partial x$
 - B. $\partial F/\partial y$
 - C. $\partial^2 F/\partial x^2$
 - D. Both equally
18. If Canny thresholds $T_{low}=T_{high}$:
- A. Optimal edge map
 - B. Degenerates to simple thresholding — hysteresis has no effect
 - C. All pixels become edges
 - D. No edges detected
19. $\nabla^2 f>0$ at a pixel indicates:
- A. Local intensity maximum
 - B. Local intensity minimum — pixel darker than its surroundings
 - C. An edge boundary
 - D. Uniform intensity region
20. The LoG computed as $(\nabla^2 G)*F$ instead of $\nabla^2(G*F)$: these are equal because:
- A. Only for symmetric kernels
 - B. By associativity of convolution: $\nabla^2(G*F)=(\nabla^2 G)*F$ — exact equality, not approximation
 - C. Only when image has zero mean
 - D. Approximate for small σ only

T3: FITTING: HOUGH & RANSAC

1. Why use normal form $p=x\theta+y\theta$ instead of slope-intercept $y=mx+b$?
- A. Fewer accumulator cells
 - B. Slope m is unbounded for vertical lines; $p\in(-D,D)$ and $\theta\in[0,\pi)$ are always bounded
 - C. Normal form more accurate for diagonal lines
 - D. Slope-intercept doesn't support voting
2. A single image point (x_0,y_0) maps to in Hough (p,θ) space:
- A. A single point (p_0,θ_0)
 - B. A sinusoidal curve $p=x_0\cos\theta+y_0\sin\theta$
 - C. A horizontal line at $p=x_0$
 - D. A circle centered at (x_0,y_0)
3. Two collinear points produce two sinusoids in Hough space. Where do they intersect?
- A. At origin $(p=0,\theta=0)$
 - B. At (p,θ) corresponding to the line through both points
 - C. They don't intersect if the line is diagonal
 - D. At two points, creating ambiguity
4. Required RANSAC iterations n for probability p with outlier fraction ϵ and sample size s :
- A. $n=((1-p))/(\epsilon^s)$
 - B. $n=((1-p))/(1-(1-\epsilon)^s)$
 - C. $n=p/(1-\epsilon)^s$
 - D. $n=s\cdot\{1\}/1-p$
5. RANSAC inlier threshold τ determines:
- A. How many iterations to run
 - B. Which points define the minimal sample
 - C. Whether a point supports the hypothesis: $|\text{residual}|\leq\tau$
 - D. The probability of model correctness
6. Hough for circles with KNOWN radius r needs what accumulator?
- A. 1D (angle θ only)
 - B. 2D (center c_x,c_y)
 - C. 3D (c_x,c_y,r)
 - D. 4D
7. As number of model parameters increases, Hough becomes impractical because:
- A. Sinusoids become harder to compute
 - B. The accumulator grows exponentially in dimension — memory and computation scale as $O(B^d)$
 - C. More edge points required
 - D. Transform loses closed-form expression
8. After RANSAC identifies inliers, the standard final step:
- A. Accept minimal-subset model as-is
 - B. Refit using ALL inliers via least squares — much better accuracy
 - C. Run RANSAC again on the inliers
 - D. Reduce τ and re-vote

9. For fitting a homography (8 DOF) with RANSAC, minimum sample size s:
- 2
 - 4
 - 6
 - 8
10. For RANSAC line fitting (s=2) with $\epsilon=0.5$ and $p=0.99$, approximately:
- 5 iterations
 - 17 iterations
 - 100 iterations
 - 500 iterations
11. Finer Hough accumulator resolution trades off:
- Better localization at cost of more noise sensitivity and memory
 - Worse localization for faster computation
 - No effect on quality
 - Fewer votes per bin
12. RANSAC's degenerate case occurs when:
- All points are inliers
 - The minimal sample is geometrically degenerate (e.g. collinear points for a homography)
 - Inlier threshold is too large
 - s equals total point count
13. RANSAC fails when outlier fraction $\epsilon \rightarrow 0.9$ because:
- Algorithm resamples the same set
 - $(1-\epsilon)^s$ is tiny; achieving $p=0.99$ requires exponentially many iterations
 - Consensus threshold must be negative
 - Too few inliers for consensus
14. The RANSAC slides state: 'How big a subset? Smallest possible.' Why is the minimal sample size preferred?
- Larger subsets produce more accurate model fits
 - With the smallest possible subset, the probability that all selected points are inliers is highest — minimizing the required number of iterations
 - Larger subsets would violate the RANSAC convergence guarantee
 - The minimal subset is always exactly 2 points regardless of the model
15. The Hough transform slides state: 'Hardly ever satisfactory in practice, because problems with noise and cell size defeat it.' The two fundamental issues are:
- Computational cost and inability to detect circles
 - Cell size (too big: cannot distinguish lines; too small: noise causes lines to be missed) and difficulty deciding how many lines are present
 - The sinusoidal mapping is approximate and loses accuracy for near-vertical lines
 - The Hough array requires too much memory for images larger than 512×512
16. Voting in Hough is robust to outliers because:
- Uses all image pixels
 - Inliers coherently vote for the correct (p, θ) ; noise adds diffuse uncorrelated votes with low density
 - Optimizes a smooth objective
 - Requires no parameter tuning
17. Each point correspondence in homography DLT gives:
- 1 linear equation
 - 2 linear equations
 - 4 linear equations
 - 8 linear equations
18. The consensus set in RANSAC is:
- All randomly sampled minimal subsets
 - All points consistent with the current hypothesis (residual $< \tau$)
 - The parameter vector of the best model
 - The set of outliers
19. Hough peak at vote count V means:
- Exactly V edge pixels are collinear
 - Approximately V edge pixels are consistent with that (p, θ) line, within quantization error
 - V is the probability the line is correct
 - The line passes through exactly V image pixels
20. RANSAC is described as a 'hypothesize-and-test' framework. Given a dataset with N points and inlier fraction $(1-\epsilon)$, what is the expected number of inliers in the final consensus set?
- s (the minimum sample size)
 - $N(1-\epsilon)$ — approximately the total inlier count
 - (N) — grows logarithmically
 - Exactly $N/2$ regardless of ϵ

T4: SAMPLING, ALIASING & PYRAMIDS

1. Nyquist theorem: to reconstruct a signal of max frequency f_{max} without aliasing, sample at $f_{\text{s}} \geq$:
- f_{max}
 - $2f_{\text{max}}$
 - $4f_{\text{max}}$
 - Proportional to signal amplitude
2. Before $2\times$ downsampling, the anti-aliasing filter cutoff should be at:
- Original Nyquist $f_{\text{s}}/2$
 - New Nyquist $f_{\text{s}}/4$ — max frequency representable after downsampling
 - Same cutoff as output resolution
 - No filter needed
3. Aliasing in images appears as:
- Uniform blurring
 - Moiré patterns, jaggies on diagonals, or false low-frequency structures not in the original
 - Salt-and-pepper noise
 - Color channel misalignment
4. Gaussian pyramid level $G_{\{i+1\}}$ from G_i :

- Subtracting adjacent levels
 - Blurring with Gaussian then downsampling by 2 in each dimension
 - Applying Gabor filters at multiple orientations
 - Computing LoG at increasing σ
5. Laplacian pyramid level $L_i = G_i - \text{EXPAND}(G_{\{i+1\}})$ represents:
- Absolute difference between consecutive blur levels
 - Bandpass residual — detail lost going from scale i to coarser scale $i+1$
 - Edge map at scale i
 - Low-pass image at scale i
6. Reconstructing G_0 from the Laplacian pyramid:
- Sum all L_i without upsampling
 - Start from coarsest G_K ; iteratively $G_i = L_i + \text{EXPAND}(G_{\{i+1\}})$ up to G_0
 - Take Laplacian of coarsest level
 - Subtract all L_i from finest level
7. Laplacian pyramid blending avoids seams because:
- A single smooth mask at full resolution
 - Blending performed independently at each frequency band with a smooth mask — no sharp transition at any scale
 - Laplacian removes DC components causing banding
 - Each level independently normalized
8. Box filter is a poor anti-aliasing filter because:
- Not separable
 - Its DFT is a sinc — sidelobes pass energy well above the intended cutoff
 - Amplifies near-cutoff frequencies by $2\times$
 - Cannot be implemented with integers
9. Total pixels in a K-level Gaussian pyramid of $N \times N$ image:
- $K \cdot N^2$
 - N^2 unchanged
 - $(4/3)N^2$ — about 33% overhead
 - N^2/K
10. EXPAND step multiplies kernel by 4 because:
- Kernel must be enlarged by 4
 - Inserting zeros reduces mean by $4\times$; multiplying by 4 restores correct amplitude
 - Anti-aliasing requires $4\times$ larger kernel
 - Nyquist frequency doubles after upsampling
11. Gabor filters are used for texture analysis because they:
- Detect only H or V patterns
 - Are separable and cheap to compute
 - Respond to oriented sinusoidal patterns at specific spatial frequency and orientation — like V1 simple cells
 - Are scale- and rotation-invariant
12. A signal at $f_{\text{s}}=150$ Hz with $f_{\text{max}}=100$ Hz aliases to:
- 50 Hz
 - 100 Hz
 - 150 Hz
 - No aliasing at 150 Hz
13. The Nyquist frequency for sampling rate f_{s} :
- f_{s}
 - $f_{\text{s}}/2$
 - $2f_{\text{s}}$
 - $f_{\text{s}}/N(2)$
14. Gaussian semigroup property $G(\sigma_1) * G(\sigma_2) = G(\sqrt{\sigma_1^2 + \sigma_2^2})$ is important for pyramids because:
- Enables integer arithmetic
 - Cascading Gaussian blurs = a single larger Gaussian — mathematically consistent multi-scale representation
 - Minimizes the LoG at each level
 - Unique to Gaussian among all kernels

15. Laplacian pyramid is useful for image compression because:
- Each level equals the original — no redundancy
 - Laplacian residuals are near-zero in smooth regions — sparse, compressible with few bits
 - Coarsest Gaussian contains all information
 - Equivalent to run-length encoding
16. Mipmaps in 3D graphics are essentially:
- Laplacian pyramids stored with texture data
 - Pre-computed Gaussian pyramids of textures — avoiding aliasing when surfaces are distant and map to few pixels
 - Anti-aliasing filters for polygon edges
 - Compressed normal maps
17. Supersampling/MSAA anti-aliases by:
- Increasing polygon count
 - Rendering at higher resolution then downsampling (averaging) — implementing low-pass filtering before display
 - Applying post-process sharpening
 - Using mipmaps only

18. A checkerboard with 2-pixel-wide squares has spatial frequency:
- 1 cycle/pixel
 - 0.5 cycles/pixel
 - 2 cycles/pixel
 - 0.25 cycles/pixel
19. Increasing σ of pre-blur Gaussian before $2\times$ downsampling:
- Increases aliasing
 - More aggressively suppresses frequencies $> f_{\text{s}}/4$ — reduces aliasing more but also loses more fine detail
 - Has no effect on aliasing
 - Increases effective sampling rate
20. The Laplacian pyramid is called 'Laplacian' because:
- Each level uses ∇^2 for edge detection
 - Each band approximates the Laplacian (second derivative) of the Gaussian-smoothed image at the corresponding scale
 - Reconstruction uses the Laplacian inverse
 - Named after Laplace who invented the transform

T5: DEEP LEARNING FUNDAMENTALS

1. Sigmoid derivative: $\sigma'(x) = \sigma(x)(1-\sigma(x)) \leq 0.25$. After L layers gradient multiplied by at most:
- 0.25L
 - 0.25^L — exponentially vanishing
 - $L^L \cdot 2$
 - $1-0.25L$
2. ReLU $f(x) = (0, x)$ is preferred over sigmoid for deep networks because:
- ReLU is bounded, preventing overflow
 - Gradient =1 for $x>0$ — no saturation, no vanishing gradient; computationally trivial
 - ReLU output follows Gaussian distribution
 - ReLU is differentiable everywhere
3. Backpropagation computes:
- Forward-pass activations for detecting active neurons
 - $\partial L / \partial w$ for all parameters via chain rule — enabling gradient descent updates
 - The Hessian of the loss
 - The Jacobian of output with respect to input
4. Batch normalization at test time uses:
- Test batch statistics
 - Running statistics (EMA of μ and σ^2) accumulated during training — fixed at inference
 - No normalization at test time
 - Global training-set statistics computed once
5. Dropout: neuron kept with probability p. At test time:
- Dropout still applied
 - All units active; weights multiplied by p (or inverted dropout: activations divided by p during training)
 - Units permanently pruned by dropout frequency
 - Output averaged over multiple stochastic forward passes
6. Zero weight initialization causes:
- Slow but eventually correct convergence
 - All neurons compute identical functions — symmetry never breaks; all neurons in a layer remain identical forever
 - Numerical overflow
 - Convergence to global minimum directly
7. Adam optimizer combines:
- Nesterov momentum only
 - First moment $m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t$ (momentum) and second moment $v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$ (adaptive LR) with bias correction
 - Gradient clipping and weight decay
 - Cyclical LR and batch normalization
8. L2 regularization: the gradient update for w_i :
- $w_i \leftarrow w_i - \partial L / \partial w_i$ unchanged
 - $w_i \leftarrow w_i (1-2\lambda) - \partial L / \partial w_i$ — multiplicative shrinkage each step
 - $w_i \leftarrow w_i - \partial L / \partial w_i + \lambda / |w_i|$
 - $w_i \leftarrow w_i - \lambda$ constant subtraction
9. Cross-entropy $L = -\sum_k y_k \log p_k$ preferred over MSE for classification because:
- Cross-entropy always smaller
 - Convex only for binary tasks
 - With softmax: gradient $\partial L / \partial z_k = p_k - y_k$ — proportional to prediction error, large when badly wrong
 - MSE cannot handle probabilities
10. 'Dead ReLU' problem and fix:
- ReLU with gradient =1 slow training; use clipped gradients
 - Permanently negative pre-activations zero output, zero gradient. Fix: Leaky ReLU $f(x) = (\alpha x, x)$, $\alpha=0.01$
 - ReLU networks can only represent convex functions
 - Dead ReLUs appear only in output layer
11. BatchNorm normalizes: $z^{\text{d}}_d = (z_d - \mu_d) / \sqrt{\sigma_d^2 + \epsilon}$, then:
- z^{d}_d is the final output
 - $y_d = \gamma_d z^{\text{d}}_d + \beta_d$ — learned scale γ_d and shift β_d restore representational capacity
 - The mean μ_d is a learned parameter
 - BN applies only to the final layer
12. Momentum $v_t = \gamma v_{t-1} + \nabla L$, $w \leftarrow w - v_t$ accelerates training by:
- Eliminating need for LR tuning
 - Accumulating consistent gradient direction; dampening oscillations perpendicular to the optimum
 - Providing unbiased gradient estimate
 - Guaranteeing escape from saddle points
13. Softmax with temperature T: $p_i = (z_i/T) / \sum_j (z_j/T)$. As $T \rightarrow 0$:
- Uniform $p_i = 1/K$
 - One-hot (argmax) — distribution sharpens to a single class
 - No effect
 - Selects top-2 classes
14. In a multi-layer perceptron, the training objective is $\theta^* = \arg\min_{\theta} \mathcal{L}(\{x, y\})$ ($y, \arg\min_{\theta} \mathcal{L}(\{x, \theta_{[1..L]}\})$). Computing $\partial \mathcal{L} / \partial \theta_i$ for early layers requires:
- Computing the Hessian of L with respect to all parameters simultaneously
 - Applying the chain rule backward from the output layer, accumulating gradient terms through each layer
 - Running a separate forward pass for each parameter
 - The gradient only depends on the immediately adjacent layers — no chain rule needed
15. SGD batch size 1 vs. full-batch gradient descent:
- Accuracy vs. memory
 - Noisy-but-cheap updates (frequent) vs. accurate gradient (slow per epoch)
 - Convergence guarantee vs. none
 - L1 vs. L2 regularization
16. Overfitting is characterized by:
- High training error and high validation error

- B. Low training error, high validation error — model memorizes training data
 - C. Slow convergence
 - D. Identical train and validation error
17. Internal covariate shift (addressed by BatchNorm):
- A. Mismatch between training and test distributions
 - B. Changing distribution of layer inputs during training as preceding layers update — each layer must constantly re-adapt
 - C. Difference between batch and population statistics
 - D. Gradient magnitude differences between layers
18. Learning rate scheduling (cosine annealing) is beneficial because:
- A. Constant large LR always converges faster
 - B. Large initial LR explores broadly; decaying LR allows fine convergence near the optimum
 - C. Eliminates need for momentum
 - D. Only needed for small datasets
19. A fully connected network $784 \rightarrow 256 \rightarrow 128 \rightarrow 10$. Total parameters (weights + biases):
- A. 236,1288
 - B. 235,146
 - C. 3,534
 - D. 1,178
20. Perceptron convergence theorem guarantees convergence only when:
- A. Activation is continuous
 - B. Data is linearly separable
 - C. Learning rate is sufficiently small
 - D. Training examples exceed the weight count

T6: CNN ARCHITECTURES

1. Output size of a conv layer: input W , kernel k , padding p , stride s :
- A. $(W-k+p)/s$
 - B. $(W-k+2p)/s+1$
 - C. $(W+2p-k) \times s$
 - D. $W/s+k$
2. ResNet skip connections $H(x)=F(x)+x$ solve the degradation problem by:
- A. Using ReLU instead of sigmoid
 - B. Providing direct gradient path: $\{\partial L\} \{ \partial x \} = \{\partial L\} \{ \partial H(1 + \partial F) \{ \partial x \} \}$ — gradient always flows
 - C. Using BN after every layer
 - D. Geometrically increasing channels
3. ResNet bottleneck: $1 \times 1 \rightarrow 3 \times 3 \rightarrow 1 \times 1$. The 1×1 convolutions:
- A. Apply BN between 3×3 layers
 - B. Reduce channels before expensive 3×3 , then expand back — same output width, much less computation
 - C. Implement the skip connection
 - D. Improve translation equivariance
4. Two 3×3 vs one 5×5 conv: same RF, but two 3×3 preferred because:
- A. 3×3 uses integer arithmetic; 5×5 does not
 - B. Two 3×3 : $2 \times 9C^2 = 18C^2$ params vs $25C^2$ for one 5×5 — fewer params, more nonlinearities
 - C. Two 3×3 are faster due to memory alignment
 - D. 5×5 cannot be used with BN
5. Weights (no bias) in conv layer $C_{in}=64, C_{out}=128, k=3$:
- A. $64 \times 128 = 8,192$
 - B. $3 \times 3 \times 64 \times 128 = 73,728$
 - C. $3 \times 3 \times 128 = 1,152$
 - D. $64 \times 128 \times 3 = 24,576$
6. Global Average Pooling (GAP) before a classifier:
- A. Computes mean across channels per spatial location
 - B. Averages each feature map spatially $\rightarrow$ one value per channel, replacing large FC layers
 - C. Takes max across all feature maps globally
 - D. Averages the conv layer weights
7. In a convolutional layer with stride $s > 1$, the output spatial size compared to stride $s = 1$ (same kernel size and padding):
- A. Increases by factor s
 - B. Decreases by approximately factor s — stride controls the downsampling rate
 - C. Stays the same — stride only affects computation, not output size
 - D. Decreases by factor s^2
8. In a CNN, applying a 1×1 convolution with $C_{out} < C_{in}$ filters before a 3×3 convolution:
- A. Increases the spatial receptive field before the 3×3
 - B. Reduces the number of channels — making the 3×3 computation cheaper while preserving the overall spatial resolution
 - C. Has no effect on computation since 1×1 convolutions are free
 - D. Acts as a max-pooling operation across channels
9. Receptive field after n stride-1 3×3 conv layers (no pooling):
- A. $3n \times 3n$
 - B. $(2n+1) \times (2n+1)$
 - C. $3^n \times 3^n$
 - D. $n \times n$
10. Transposed convolution in decoder networks:
- A. Removes conv artifacts
 - B. Learned upsampling: inserts zeros between input pixels, applies a learned filter — increases spatial resolution
 - C. Exactly inverts a previous conv layer
 - D. Fixed bicubic upsampling
11. BatchNorm with batch size $B=1$ is problematic because:
- A. Batch mean is zero for any single sample
 - B. Batch variance $= (1/B) \sum (x_i - \mu)^2 = 0$ for $B=1$ — normalization becomes degenerate
 - C. Learned γ, β cannot update with one sample
 - D. BN with $B=1$ equals no normalization

12. ResNet skip connections add the input x directly to the output: $H(x)=F(x)+x$. If the network is initialized near zero ($F(x)=0$), then at initialization:
- A. $H(x)=0$ — the network outputs near-zero initially
 - B. $H(x)=x$ — the skip connection provides an identity mapping, giving a sensible initialization
 - C. $H(x)=2x$ — the addition doubles the signal
 - D. The skip connection is disabled at initialization and only activates after training
13. Deeper layers in a CNN have larger receptive fields but lower spatial resolution. This creates a tradeoff because:
- A. Larger receptive fields reduce translation equivariance
 - B. Deeper layers capture more semantic/global context but lose precise spatial localization that shallow layers retain
 - C. Deeper layers are computationally cheaper but less accurate
 - D. Larger receptive fields require more parameters per layer
14. 1×1 convolution with C_{out} filters implements:
- A. Spatial max-pooling across channels
 - B. Learned $C_{in} \times C_{out}$ linear projection of the feature vector at each spatial location
 - C. Spatial gradient across channels
 - D. Channel-wise independent scaling (diagonal matrix)
15. Strided convolution vs max-pooling for downsampling:
- A. Strided conv is always faster
 - B. Strided conv learns the downsampling kernel; max-pooling uses a fixed rule (take maximum)
 - C. Max-pooling is differentiable; strided conv is not
 - D. Strided conv can only halve resolution
16. Degradation problem (deep plain network trains worse than shallow) is not overfitting because:
- A. Training error (not just test error) ALSO increases with depth
 - B. Dataset is too small
 - C. SGD cannot optimize deep networks
 - D. Too many parameters
17. Max-pooling and average-pooling both downsample feature maps. The key difference:
- A. Max-pooling is differentiable; average-pooling is not
 - B. Max-pooling selects the most strongly activated feature in each region (good for detecting presence); average-pooling summarizes the mean activation (translation invariant summary)
 - C. Max-pooling doubles the number of channels; average-pooling keeps channels fixed
 - D. Average-pooling requires larger kernel sizes to achieve the same downsampling
18. VGG's multiple 3×3 vs one large kernel:
- A. Only 3×3 kernels support BN
 - B. Same effective RF, fewer parameters ($n \times 9C^2$ vs $(2n+1)^2 C^2$), more nonlinearities
 - C. Large kernels cannot be parallelized on GPU
 - D. Single large kernel cannot be backpropagated through
19. A convolutional autoencoder with bottleneck forces:
- A. More channels than the input
 - B. Compressed representation: encoder must retain only features most useful for reconstruction
 - C. A random projection of the input
 - D. Exact inversion of each conv layer
20. 1×1 conv in ResNet (bottleneck): the 1×1 before 3×3 reduces channels to:
- A. Preserve gradient flow through the identity shortcut
 - B. Reduce the computational cost of the 3×3 conv — operating in a lower-dimensional space
 - C. Apply batch normalization more efficiently
 - D. Enable max-pooling across channels

T7: IMAGE SEGMENTATION

1. K-means minimizes $J = \sum_i \sum_{x \in C_i} \|x - c_i\|^2$. Convergence is guaranteed because:
- A. Objective is convex
 - B. E-step (assignment) and M-step (update centers) never increase J — bounded below by 0
 - C. Objective is differentiable everywhere
 - D. K-means is equivalent to Gaussian EM
2. K-means++ initialization selects centers with probability:
- A. Uniformly at random
 - B. Proportional to $D(x)^2$ — squared distance from x to nearest already-chosen center
 - C. Based on histogram peaks
 - D. By running K-means on a random subsample
3. Graph cuts for binary segmentation: the min-cut minimizes:
- A. Sum of all edge weights
 - B. Total weight of edges crossing the cut — the boundary cost of the fg/bg partition
 - C. Number of pixels in the smaller segment
 - D. Squared difference of cluster means
4. Normalized Cut (NCut) avoids trivially small segments by normalizing by:
- A. Number of pixels in each segment
 - B. Total association $\text{assoc}(A, V) = \sum_{\{u \in A, v \in V\}} w(u, v)$
 - C. Variance of pixel intensities within each segment
 - D. Boundary length divided by area
5. NCut is solved via:
- A. SVD of affinity matrix W
 - B. Generalized eigenvalue problem $(D-W)v = \lambda Dv$; the Fiedler vector (2nd smallest eigenvector) gives the bipartition
 - C. Power iteration on W
 - D. Sparse Cholesky factorization
6. Snake total energy: $E = \epsilon \{ \alpha \|v - s\|^2 + \beta \|v_{ss}\|^2 \} ds + E_{\text{ext}}$. The $\beta \|v_{ss}\|^2$ term controls:
- A. Attraction to image edges
 - B. Bending resistance — penalizes curvature/sharp corners, enforcing smooth curves
 - C. Contour length
 - D. Area enclosed by contour
7. External energy $E_{\text{ext}} = -\int I(v(s)) \|v'\|^2 ds$ drives the snake toward:
- A. Bright regions
 - B. High-gradient regions (edges) — minimizing negative gradient means maximizing $|\nabla I|$
 - C. Uniform color regions
 - D. Corners only
8. Mean shift segmentation finds:
- A. Cluster centers by random sampling + K-means
 - B. Modes of the feature-space density by iteratively shifting each point to the local weighted mean of neighbors
 - C. Global minimum of a convex objective
 - D. Watershed lines in the gradient image
9. A snake's main limitation:
- A. Can only segment circles
 - B. Requires initialization close to the boundary — far initializations see negligible gradient and cannot reach the target
 - C. Cannot handle more than 2 objects
 - D. Works only on binary images
10. The slides note that K-means 'doesn't answer basic questions like: what's the probability that a point x is in cluster m ?' This motivates:
- A. Graph-based segmentation methods
 - B. Probabilistic clustering approaches (e.g. mixture models) that model the full distribution over cluster assignments
 - C. Increasing K until each cluster has exactly one point
 - D. Using the Hough transform instead of K-means
11. Semantic vs. instance segmentation:
- A. Semantic: pixel $\rightarrow$ probability; Instance: pixel $\rightarrow$ hard label
 - B. Semantic: pixel $\rightarrow$ class (no instance distinction); Instance: pixel $\rightarrow$ (class, instance ID)
 - C. Semantic: only one class; Instance: multiple classes
 - D. Semantic is coarser spatially
12. The slides show three types of segmentation: semantic, instance, and panoptic. Panoptic segmentation:
- A. Is the same as semantic segmentation but applied to medical images
 - B. Combines semantic labeling (class per pixel) with instance labeling (individual object IDs), evaluated by Panoptic Quality = Segmentation Quality $\times$ Recognition Quality
 - C. Only segments foreground objects, ignoring background
 - D. Requires a stereo camera setup to separate instances by depth
13. In histogram-based color segmentation, the goal is to reduce the image to K colors and assign each pixel to its closest color. The slides show that with $K=2$:
- A. The image is converted to grayscale
 - B. Each pixel is mapped to one of two representative colors — producing a binary segmentation based on the dominant colors
 - C. The image is split into foreground and background based on depth
 - D. Two Gaussian distributions are fit to the color histogram using EM
14. The slides state that deformable contours (snakes) are 'initialized near the desired object boundary.' What happens if the initial contour is placed far from the true boundary?
- A. The snake will always converge to the correct boundary given enough iterations
 - B. The image energy gradient ∇E_{ext} is small far from edges — the snake sees no force toward the boundary and remains stuck
 - C. The internal energy term automatically moves the snake toward the nearest edge
 - D. The snake contracts uniformly until it finds an edge
15. Graph cut data term $D(p, fg)$ represents:
- A. Boundary penalty between neighboring pixels
 - B. Cost of assigning p to foreground: $-P(p|fg \text{ model})$ — penalizes assignments inconsistent with appearance model
 - C. Distance from p to nearest edge
 - D. Minimum cut cost for the full graph
16. The slides define the cost of a graph cut as $\text{cut}(A, B) = \sum_{\{u \in A, v \in B\}} w(u, v)$. A low-cost cut tends to pass through:
- A. Regions where pixel intensities are very similar
 - B. Regions where adjacent pixels have very different intensities (edges) — where similarity weights $w(u, v)$ are small
 - C. The geometric center of the image
 - D. Regions with many pixels — ensuring balanced segment sizes
17. K-means is NOT guaranteed to find the global minimum because:
- A. E-step is nonconvex
 - B. The objective has many local minima; algorithm converges to the nearest local minimum from initialization
 - C. Floating-point M-step introduces errors
 - D. The objective is non-differentiable
18. Smoothness term $\sum_{\{(p, q) \in N\}} w_{pq} \cdot \{1 - |p - q|\}$:
- A. Penalizes pixels far from foreground centroid
 - B. Penalizes adjacent pixels with different labels, weighted by similarity: w_{pq} small at edges (cheap to cut), large in smooth regions
 - C. Penalizes total number of background pixels
 - D. Penalizes pixels with ambiguous appearance
19. Snake elasticity term $\alpha \|v - s\|^2$ ($v, s = \partial v / \partial s$) controls:
- A. Bending resistance
 - B. Elasticity — penalizes stretching/lengthening, acting like rubber band tension
 - C. External gradient attraction
 - D. Area enclosed by contour
20. Histogram-based color quantization assigning each pixel to nearest cluster is the K-means:
- A. M-step (update cluster centers)
 - B. E-step (assign each pixel to nearest center)
 - C. K-means++ initialization
 - D. Convergence check

T8: MOTION ESTIMATION & OPTICAL FLOW

1. The optical flow constraint $I_x u + I_y v + I_t = 0$ is derived from:
- A. Spatial smoothness of the image
 - B. Brightness constancy $I(x,y,t) = I(x+u,y+v,t+1)$ — first-order Taylor expansion
 - C. Purely translational motion assumption
 - D. Fixed depth scene assumption
2. The aperture problem states:
- A. Large displacements cannot be computed
 - B. $I_x u + I_y v + I_t = 0$ has two unknowns — only the normal-to-edge component of flow is observable from a single pixel
 - C. Flow fails with multiple moving objects
 - D. Temporal aliasing prevents flow at high frame rates
3. Lucas-Kanade (LK) assumes constant flow in window, solving $A^T A(u) = A^T T(b)$ where $A^T A =$:
- A. $\sum I_x^2 \ I_x \ I_y \ I_x \ I_y \ I_y^2$ (structure tensor)
 - B. $\sum I_x \ 0 \ 0 \ 0 \ \sum I_y$
 - C. $\sum I_x^2 \ 0 \ 0 \ \sum I_y^2$
 - D. $(\sum I_x^2 + \sum I_y^2)I$
4. LK fails when $A^T A$ is:
- A. Positive definite with two large eigenvalues (corner)
 - B. Near-singular (small eigenvalues) — uniform regions or straight edges: aperture problem persists
 - C. Diagonal
 - D. Computed from only 2 pixels
5. The slides outline motion estimation as: (1) patch-based, (2) image warping, (3) parametric (global) motion. What distinguishes parametric global motion estimation from patch-based Lucas-Kanade?
- A. Global methods assume each pixel has an independent velocity
 - B. Global methods estimate a single parametric model (e.g. affine: 6 parameters) that describes the motion of the entire image — more robust but less flexible than per-pixel flow
 - C. Global methods require a stereo camera setup
 - D. Global methods solve for optical flow using only temporal differences, not spatial gradients
6. Hierarchical coarse-to-fine optical flow solves:
- A. The aperture problem at every scale
 - B. Large displacements: at coarser scale effective displacement is small (within Taylor expansion validity); refined at finer scales
 - C. Horn-Schunck local minima
 - D. Brightness constancy failure under color changes
7. The affine motion model $(x',y') = (ax+by+c, dx+ey+f)$ has how many parameters?
- A. 4
 - B. 6
 - C. 8
 - D. 9
8. Block matching motion estimation (used in MPEG/H.264):
- A. Computes dense optical flow then quantizes to blocks
 - B. For each $N \times N$ block, finds best match in reference frame within $\pm d$ search window by SAD or SSD
 - C. Fits affine model to each block independently
 - D. Applies LK with window equal to block size
9. Inverse (backward) warping is preferred over forward warping because:
- A. More accurate for large rotations
 - B. For each output pixel, maps back to a source position and interpolates — no holes or overlaps
 - C. Better photometric consistency
 - D. Less memory required
10. Brightness constancy fails under:
- A. Displacements larger than 1 pixel
 - B. Changing illumination, moving specular highlights, shadows, motion blur — change pixel brightness independent of motion
 - C. JPEG compression
 - D. Multiple moving objects
11. KLT tracker selects features using:
- A. Maximum gradient magnitude
 - B. $(\lambda_1, \lambda_2) > \tau$ — max of the smaller eigenvalue of $A^T A$ (Shi-Tomasi criterion)
 - C. Random selection followed by flow verification
 - D. Points on detected Canny edges
12. Bilinear interpolation at non-integer position $(x+\alpha, y+\beta)$:
- A. $I[x,y]$ (nearest neighbor)
 - B. $(1-\alpha)(1-\beta)I[x,y] + \alpha(1-\beta)I[x+1,y] + (1-\alpha)\beta I[x,y+1] + \alpha\beta I[x+1,y+1]$
 - C. 4x4 cubic spline
 - D. Gaussian-weighted average within radius 1
13. For a horizontal edge ($I_y=0, I_x \neq 0$), the optical flow constraint gives:
- A. u fully determined; v unconstrained
 - B. v fully determined; u unconstrained
 - C. Both u and v determined
 - D. Neither determined
14. SSD for block matching at displacement (u,v) :
- A. $\sum_{(x,y) \in \text{block}} (I_1(x,y) - I_2(x+u,y+v))^2$
 - B. $\sum (I_1 - I_2)$
 - C. $|I_1 - I_2|$
 - D. $(\sum |I_1 - I_2|) / (W^2)$ normalized
15. Optical flow CANNOT directly recover, without additional constraints:
- A. Temporal frame interpolation
 - B. Action recognition from motion patterns
 - C. 3D scene shape from 2D flow alone
 - D. Video stabilization
16. KLT tracker fails when:
- A. Tracked point is at a corner
 - B. Point moves out of frame, becomes occluded, or displacement exceeds the pyramid range
 - C. Image has too many texture points

- D. Frame rate is too high
17. A homography motion model is EXACT when:
- A. Camera has any motion with general 3D scene
 - B. Scene is planar (any camera motion) OR camera rotates/zooms with no translation (any 3D scene)
 - C. Camera translates purely horizontally
 - D. Scene has affine structure
18. Image stabilization using motion estimation:
- A. Increases frame rate to reduce blur
 - B. Estimates global camera motion between frames; applies the inverse transform to remove camera shake
 - C. Applies temporal Gaussian smoothing to each pixel track
 - D. Predicts future frames to fill blurred regions
19. Hierarchical LK: after estimating flow u^i, v^i at level i :
- A. Simply add u^i, v^i to all previous estimates
 - B. Warp second image by u^i, v^i , upsample estimate $\times 2$ as initial guess for level $i+1$, estimate residual
 - C. Re-run LK on full-resolution image
 - D. Discard estimate and restart

T9: TRACKING & PARTICLE FILTERS

1. Kalman filter is optimal (minimum MSE) under:
- A. Uniform prior
 - B. Linear dynamics $x_t = Ax_{t-1} + w$ ($w(0,Q)$) and linear measurements $z_t = Hx_t + v$ ($v(0,R)$)
 - C. Discrete finite state space
 - D. Measurements exceeding state dimension
2. The Kalman filter PREDICT step:
- A. $x_t^{\wedge} \{t\} = x^{\wedge}_{t-1}$; $P_t \{t\} = P_{t-1}$
 - B. $x_t^{\wedge} \{t-1\} = Ax^{\wedge}_{t-1} \{t-1\}$; $P_t \{t-1\} = AP_{t-1} \{t-1\} A^T + Q$
 - C. $x_t^{\wedge} \{t\} = Hx^{\wedge}_{t-1} \{t-1\}$; $P_t \{t-1\} = HP_{t-1} \{t-1\} H^T + R$
 - D. $x_t^{\wedge} \{t-1\} = A^{\wedge} \{t-1\} x^{\wedge}_{t-1}$; $P_t \{t-1\} = P_{t-1} - Q$
3. The Kalman gain K_t :
- A. $K_t = P_t \{t-1\} H^T (HP_t \{t-1\} H^T + R)^{-1}$
 - B. $K_t = Q(Q+R)^{-1}$
 - C. $K_t = H^T \{t-1\}$
 - D. $K_t = R/(P+R)$
4. Particle filter represents $p(x_t | z_{1:t})$ as:
- A. Single Gaussian $N(\mu, \Sigma)$
 - B. Weighted set $\{x_t^{\wedge} \{i\}, w_t^{\wedge} \{i\}\}_{i=1}^N$ of N particles
 - C. Fixed K -component Gaussian mixture
 - D. Grid discretization of state space
5. Low effective sample size $N_{\text{eff}} = 1/\sum_i (w_i^{\wedge})^2 \approx 1$ indicates:
- A. Well-spread equally-weighted particles
 - B. Particle degeneracy: one particle has weight ≈ 1 , all others negligible — distribution collapsed
 - C. Very accurate measurement model
 - D. High process noise Q
6. CONDENSATION particle propagation samples:
- A. All particles to the MAP estimate
 - B. $x_t^{\wedge} \{i\} p(x_t | x_{t-1}^{\wedge} \{i\})$ — the transition prior (stochastic dynamics)
 - C. Particles deterministically via Kalman prediction
 - D. Gaussian jitter with fixed $\sigma=1$
7. Systematic resampling is preferred over multinomial because:
- A. Cannot handle discrete distributions
 - B. $O(1)$ variance vs $O(N)$ for multinomial — deterministic stratified spacing covers all weight intervals
 - C. Changes total particle weight
 - D. Requires sorted particles
8. Particle filters represent multi-modal posteriors; KF cannot because:
- A. PFs use more observations per step
 - B. Particles can cluster around multiple modes simultaneously; KF maintains a single Gaussian (unimodal)
 - C. PFs have a stronger motion model
 - D. KF uses more memory
9. Measurement weighting $w_t^{\wedge} \{i\} p(z_t | x_t^{\wedge} \{i\})$ computes:
- A. Prior probability of particle state
 - B. Likelihood: how probable is observation z_t given particle's hypothesized state $x_t^{\wedge} \{i\}$
 - C. Transition probability from x_{t-1} to $x_t^{\wedge} \{i\}$
 - D. The posterior probability (Bayes-inverted)
10. Required particle count scales with state dimensionality as:
- A. $O(d)$ linear
 - B. $O(d^2)$ quadratic
 - C. Exponential in d (curse of dimensionality)
 - D. Constant — independent of d
11. The Kalman filter assumes linear Gaussian dynamics. In visual tracking, which real scenario most directly violates this assumption?
- A. The tracked object has constant velocity
 - B. An object that rotates, causing its appearance to change nonlinearly, or an object that moves out of view (creating a multimodal distribution over location)
 - C. The camera has a fixed focal length
 - D. The measurement noise covariance R is diagonal
12. In the particle filter, after propagating particles using the motion model, what must be done before resampling?
- A. Sort particles by their state values
 - B. Weight each particle by its measurement likelihood $w^{\wedge} \{i\} p(z_t | x_t^{\wedge} \{i\})$ — how well each particle explains the current observation
 - C. Reset all particle weights to $1/N$
 - D. Merge particles that are within a threshold distance of each other
13. The slides describe CONDENSATION as suitable for tracking in clutter. Why is a standard Kalman filter insufficient for cluttered scenes?
- A. The Kalman filter cannot run in real time on cluttered images

- B. Clutter creates multiple competing hypotheses (multimodal posterior) — the Kalman filter collapses this to a single Gaussian, losing all but one hypothesis
 - C. The Kalman filter requires a known object template; clutter introduces unknown templates
 - D. The Kalman filter only works for 1D state spaces
14. CONDENSATION was designed for tracking contours in clutter because:
- A. Fewer particles than generic PF
 - B. Cluttered environments create multi-modal posteriors; single-estimate trackers fail under occlusion; CONDENSATION maintains the full distribution
 - C. Contour shapes modeled as Gaussian
 - D. CONDENSATION forces contour onto image gradient
15. Kalman filter for constant-velocity tracking: state $x = [p, v]^T$, motion matrix A for Δt :
- A. $A = [1 \ 0 \ 0 \ 1]$
 - B. $A = [1 \ \Delta t \ 0 \ 1]$
 - C. $A = [\Delta t \ 0 \ 0 \ \Delta t]$
 - D. $A = [0 \ 1 \ 1 \ 0]$
16. Particle impoverishment (sample poverty) is addressed by:
- A. Dramatically increasing N
 - B. Roughening: adding small random perturbation after resampling, or MCMC moves to diversify particles
 - C. Decreasing τ in the measurement model
 - D. Keeping low-weight particles after resampling
17. SIR particle filter step order:
- A. Resample → Weight → Propagate
 - B. Weight → Propagate → Resample
 - C. Propagate → Weight → Resample (if $N_{\text{eff}} < N_{\text{thresh}}$)
 - D. Resample → Propagate → Weight
18. Kalman gain $K \rightarrow 0$ means the filter:
- A. Fully trusts the new measurement
 - B. Ignores the measurement — prediction dominates because R $HP_t \{t-1\} H^T$
 - C. State estimate is diverging
 - D. Has converged to the true state permanently
19. Bootstrap particle filter uses $p(x_t | x_{t-1})$ as proposal. The optimal alternative is:
- A. $p(z_t | x_t) = 1$ — the likelihood
 - B. $p(x_t | x_{t-1}, z_{1:t}) p(z_t | x_t) p(x_t | x_{t-1})$ — the optimal proposal incorporating current measurement
 - C. Uniform distribution over state space
 - D. Posterior from previous step
20. After resampling in a particle filter, the weights should be set to:
- A. Their previous values
 - B. $1/N$ — uniform weights, since resampling already incorporated the previous weights
 - C. Zero for low-weight particles, one for others
 - D. Proportional to their previous weights squared

T10: CAMERA MODELS & PROJECTIVE GEOMETRY

1. Pinhole projection of 3D camera-frame point (X,Y,Z) to image (x,y) :
- A. $x = X/f, y = Y/f$
 - B. $x = fX/Z, y = fY/Z$
 - C. $x = fX, y = fY$
 - D. $x = X+c, x_y = Y+c_y$
2. Camera intrinsic matrix $K = [f \ x_c \ s \ 0 \ x_0 \ f \ y_c \ 0 \ 0 \ 1]$. Skew $s \neq 0$ indicates:
- A. Different pixel sizes in x and y
 - B. Pixel axes are not orthogonal — sensor rows and columns not perpendicular
 - C. Radial distortion present
 - D. Principal point is offset
3. Full projection matrix $P = K[R|t]$ (with $s=0$) has how many DOF?
- A. 9
 - B. 10
 - C. 11
 - D. 12
4. Converting homogeneous (w_1, w_2, w_3) to Cartesian:
- A. Drop the third component: (w_1, w_2)
 - B. Divide by third component: $(w_1/w_3, w_2/w_3)$
 - C. Multiply by third component
 - D. Normalize to unit length
5. Barrel distortion ($k_1 < 0$) causes:
- A. Image corners brighter
 - B. Straight lines away from center curve inward — image appears to bulge outward at center
 - C. Pincushion: image pinches inward
 - D. Loss of border resolution
6. Zhang's checkerboard calibration method requires:
- A. A calibrated ruler at known 3D position
 - B. A planar pattern viewed in ≥ 3 different orientations
 - C. A 3D calibration rig with known 3D points
 - D. Multiple cameras observing the same scene
7. The principal point (c_x, c_y) :
- A. Always the geometric image center ($W/2, H/2$)
 - B. Image coordinates where the optical axis intersects the image plane — may differ from image center
 - C. Vanishing point for vertical lines
 - D. Camera center coordinates in world space
8. Vanishing point of 3D direction $\{d\}$:
- A. $V = K\{d\}$
 - B. $V = K[R|t] \cdot \{d\} \cdot 0^T$ — treating direction as point at infinity
 - C. $V = K^{\wedge} \{t\} \{d\}$
 - D. $V = R^{\wedge} T K \{d\}$
9. DLT for estimating P (11 DOF) requires minimum how many 3D-2D correspondences?

- 4
 - 6
 - 8
 - 11
10. Back-projecting image point (u,v) to 3D gives:
- A unique 3D point at depth $Z=f$
 - A ray: $\{C+\lambda \cdot K^{\wedge}\{-1\}(u,v,1)^{\wedge}T;\lambda>0\}$ — all 3D points projecting to (u,v)
 - The camera optical center
 - A circle at distance f
11. Without homogeneous coordinates, perspective projection is nonlinear because:
- Division by Z cannot be expressed as a matrix multiplication
 - Perspective requires spherical trigonometry
 - The image plane is curved
 - Focal length introduces a quadratic term
12. Increasing focal length f (fixed sensor size) causes:
- Wider field of view
 - Narrower FOV (telephoto), objects appear magnified, shallower depth of field
 - Same FOV, higher resolution
 - Wider FOV, larger circle of confusion
13. The circle of confusion (CoC):
- Circular image of a point at the focal distance
 - Blurred disk formed by a point NOT at focal distance; $\text{DOF} = \text{range where CoC} \leq \text{one pixel}$
 - Chromatic aberration ring
 - Fisheye lens distortion pattern
14. Extrinsic parameters $[R|t]$ encode:
- Pixel to camera coordinate mapping
 - Rigid body transform (rotation R + translation t) from world to camera coordinates
 - Lens distortion model
 - Mapping from image to 3D world points
15. Weak perspective projection is valid when:
- Camera has short focal length
 - Object depth variation δZ mean depth Z_0 AND object subtends small visual angle
 - Scene viewed from far away only
 - B conditions individually (not necessarily together)
16. Parallel lines in the ground plane share a vanishing line called:
- The epipolar line
 - The horizon line (image of the ground plane at infinity)
 - The principal line
 - The optical axis projection
17. Skew $s=0$ for modern digital cameras because:
- Short focal lengths
 - Manufacturing precision ensures pixel rows and columns are perpendicular ($\phi=90^\circ$ $s=f_x \sin \phi=0$)
 - Barrel distortion corrects skew
 - Calibration software always sets $s=0$
18. Relative rotation R_{rel} and translation t_{rel} from cam 1 to cam 2 (extrinsics $R_{1,t,1}$ and $R_{2,t,2}$):
- $R_{\text{rel}}=R_{2R,1,t,1}^{-1} R_{1,t,1}$
 - $R_{\text{rel}}=R_{2R,1,t,1}^{-1} t_{\text{rel}}=t_{2R,2R,1,t,1}^{-1} t_{1,t,1}$
 - $R_{\text{rel}}=R_{1,t,1}^{-1} t_{\text{rel}}=t_{2R,2R,1,t,1}^{-1} t_{1,t,1}$
 - $R_{\text{rel}}=R_{2R,1,t,1}$
19. Perspective foreshortening: at twice the depth, a fixed-size object appears:
- Half the size
 - One quarter the size
 - The same size
 - $1/\sqrt{2}$ the size
20. A 2D homography has how many DOF?
- 6
 - 7
 - 8
 - 9

T11: STEREO VISION & STRUCTURE FROM MOTION

1. Epipolar constraint: given point p in image 1, its correspondence in image 2 lies:
- At a unique point given by scene depth
 - On the epipolar line $l_2=FP$ — reducing 2D search to 1D
 - At the mirror image about the image center
 - In a $w \times w$ window around p
2. Essential matrix $E=[t]_{\wedge} \times R$ satisfies $\{q\}^{\wedge}TE\{p\}=0$. Derived from:
- Homogeneous coordinate multiplication
 - Coplanarity of $\{p\}, R\{q\}$, and baseline $t: \{q\}^{\wedge}T(t \times R\{p\})=\{q\}^{\wedge}T[t]_{\wedge} \times R\{p\}=0$
 - Triangle inequality in camera geometry
 - The Fundamental Theorem of Algebra
3. Fundamental matrix F vs. essential matrix E :
- F for pixel coordinates (includes K_1, K_2); E for calibrated coordinates — $F=K_2^{-1}TJ E K_1^{-1}$
 - F has rank 2; E has rank 3
 - F is symmetric; E is skew-symmetric
 - F uses 5 correspondences; E uses 8
4. The 8-point algorithm for F : the '8' refers to:
- 8 entries (the (3,3) entry fixed to 1)
 - F has 9 entries, scale-free 8 DOF; each correspondence gives 1 linear equation 8 needed
 - 8 DOF from rank-2 constraint + scale
 - 8 points needed for numerical stability only
5. The epipole e_1 in image 1:
- Projection of camera 2's optical center into image 1
 - Intersection of all epipolar lines in image 1
 - Both A and B — equivalent definitions
 - The principal point of camera 1

6. In rectified stereo with focal length f and baseline B , disparity $d=x_L-x_R$ and depth Z :
- $Z=d \cdot f/B$
 - $Z=f \cdot B/d$
 - $Z=f/(d \cdot B)$
 - $Z=B \cdot d/f$
7. Recovering R, t from E via SVD: number of candidate (R, t) pairs:
- 1 unique solution
 - 2
 - 4
 - 8
8. Bundle adjustment minimizes:
- 3D point positions only
 - Camera pose parameters only
 - $\sum_{i,j} \|\mathbf{x}_{ij} - \pi(P_{j,X_{ij}})\|^2$ — total reprojection error over all cameras and 3D points jointly
 - Feature descriptor distances
9. Hartley's normalized 8-point algorithm improves stability by:
- Using integer coordinates [0,255]
 - Normalizing correspondences to zero mean and RMS distance $\sqrt{2}$ from origin before solving
 - Independently normalizing x and y to unit variance
 - Subtracting the image center only
10. A degenerate configuration for E/F estimation:
- Many textured edges in the scene
 - All 3D points coplanar, or cameras share the same optical center ($t=0$)
 - More than 8 correspondences used
 - Low image resolution
11. Rank-2 constraint on F enforced post-hoc by:
- Dividing F by its trace
 - $F^{\wedge} \approx U \text{diag}(\sigma_1, \sigma_2, 0) V^{\wedge} T$ — setting smallest singular value to zero
 - Removing the last row of F
 - Normalizing F to unit Frobenius norm
12. Triangulation from two camera rays (noisy data):
- Exact intersection — rays always intersect in noiseless data
 - Point minimizing sum of squared distances to both rays (linear DLT or reprojection error), since noisy rays are skew
 - Midpoint of the common perpendicular
 - Projecting onto both image planes and averaging
13. PnP (Perspective-n-Point) solves for:
- Camera intrinsic matrix K
 - Camera extrinsics $[R|t]$ given known 3D points and their 2D projections
 - 3D positions of observed points
 - Fundamental matrix between two views
14. SIFT features are used in SfM because they are invariant to:
- Viewpoint changes $>45^\circ$
 - Scale, in-plane rotation, and moderate illumination changes — enabling matching across wide baseline collections
 - Motion blur and compression only
 - Color changes but not scale
15. Incremental SfM (COLMAP) builds the reconstruction by:
- Processing all images simultaneously
 - Starting with a well-matched image pair; then adding images one-by-one via PnP with periodic bundle adjustment
 - Clustering images and reconstructing clusters independently
 - Using GPS/IMU priors for every pose
16. DOF of the essential matrix E :
- 6 (3 for R + 3 for t)
 - 5 (3 for R + 2 for direction of t — scale of t is unrecoverable)
 - 4
 - 7
17. Multi-view stereo (MVS) improves over two-view stereo by:
- Using a larger baseline between exactly 2 cameras
 - Aggregating photoconsistency from many overlapping views — stronger constraint, better coverage, occlusion handling
 - Running SfM on video frames only
 - Using semantic segmentation to guide stereo

18. Stereo rectification transforms both images so that:

- Both images have the same size
- Epipolar lines become horizontal scanlines at the same y -coordinate — enabling efficient 1D correspondence search
- Lens distortion is removed before matching
- Principal points are aligned

19. Reprojection error in SfM:

- 3D distance between reconstructed and true point positions
- $\|\mathbf{x}_{ij} - \pi(P_{j,X_{ij}})\|$ — 2D distance between observed feature and projected 3D point in image j
- Descriptor distance between matched feature pairs
- Rotation angle error between estimated and true cameras

20. Planar homography H is estimated from:

- Essential matrix plus known depth
- ≥ 4 point correspondences on the planar surface
- Fundamental matrix plus known baseline
- 3 correspondences plus one distance measurement

T12: GENERATIVE MODELS

1. VAE ELBO: $L=E_{-}[q(z|x)]\{p(x|z)]-D_{KL}(q(z|x)||p(z))$. The KL term:

- Maximizes reconstruction quality
- Regularizes encoder: penalizes $q(z|x)$ for deviating from prior $p(z)=N(0,I)$ — smooths latent space for sampling
- Computes log-likelihood of data directly
- Ensures decoder is invertible

2. The reparameterization trick $z=\mu+\sigma \cdot \epsilon$, $\epsilon \sim N(0,I)$, enables:

- Closed-form KL divergence computation
- Backpropagation through sampling: $\partial z/\partial \mu=1$, $\partial z/\partial \sigma=\epsilon$ — gradient flows through the deterministic function
- Deterministic latent codes
- Sampling from bounded distributions

3. A regular autoencoder without KL regularization cannot generate effectively because:

- The bottleneck is too small
- The latent space has no enforced structure — encoder maps to disconnected, non-smooth regions; random $N(0,I)$ samples likely fall in uncovered regions
- Autoencoders cannot reconstruct accurately
- Sampling requires gradient computation

4. The slides show the Standard VAE objective as 'Reconstruction Error + KL Divergence.' What does the reconstruction error term $E_{-}[q(z|x)]\{p(x|z)]$ encourage?

- The encoder to map different inputs to the same latent code
- The decoder to reconstruct the input x accurately from the latent code z sampled by the encoder
- The latent distribution $q(z|x)$ to be close to the prior $p(z)$
- The encoder to use only a small number of latent dimensions

5. DDPM forward process: $q(x_{t|x_0})$ in closed form with $\{\alpha\}_t=\prod_{s=1}^t(1-\beta_s)$:

- $x_t=\{\alpha\}_t x_0+\{1-\alpha\}_t \epsilon, \epsilon \sim N(0,I)$
- $x_t=x_0+\sqrt{t\beta} \epsilon$
- $x_t=\beta_t x_{t-1}+(1-\beta_t) \epsilon$
- $x_t=x_0(1-\beta)^t$

6. DDPM (simplified) training objective for $\epsilon_{\theta}(x_t,t)$:

- Predict x_0 : $\|x_0-x^{\wedge}_t\|^2$
- Predict added noise: $E_{-}[x_0, \epsilon_t] \epsilon_{\theta}(\{ \alpha \}_t x_0+\{1-\alpha\}_t \epsilon, t) \|^2$
- Minimize D_{KL} at each step independently
- Maximize $p_{\theta}(x_0)$ directly

7. In the VDM formulation from the slides: 'The latent dimension is the same as the data dimension.' How does this differ from a standard VAE?

- It doesn't differ — standard VAEs also maintain the data dimension in the latent space
- Standard VAEs compress to a lower-dimensional latent; VDMs keep the latent at full data resolution and instead index latents by timestep t
- VDMs do not have a decoder; only the encoder is defined
- VDMs require the latent space to be discrete (codebook-based)

8. In the VDM forward process, $q(x_{t|x_{t-1}})$ is a linear Gaussian. The slides note that this evolves such that 'the latent at the final timestep is a standard Gaussian.' This means:

- The noise added at each step is the same fixed Gaussian
- The noise schedule $\{\beta_t\}$ is designed so $\{\alpha\}_T \rightarrow 0$, making $q(x_T)=N(0,I)$ — a known, easily sampled prior
- The forward process runs backwards from noise to data
- Only the last timestep adds noise; all previous steps are deterministic

9. The VDM ELBO (from slides) contains a reconstruction term at $t=1$. Specifically it is $E_{-}[q(x_{1|x_0})]\{p_{\theta}(x_{0|x_1})\}$. This term:

- Penalizes the encoder for making the latent x_T differ from a standard Gaussian
- Encourages the model to reconstruct the original data x_0 from a slightly noisy version x_1 at the finest scale
- Measures the consistency of forward and reverse transitions at all intermediate timesteps
- Is equivalent to the KL divergence between the prior and posterior

10. VAE ELBO reconstruction-KL tension:

- Both terms always increase together
- Maximizing reconstruction requires precise encoding (narrow posterior, high KL); minimizing KL requires diffuse posterior (poor reconstruction)
- KL term only affects the decoder
- Reconstruction maximization requires a broad, diffuse posterior

11. The VDM ELBO includes a 'prior matching' term $D_{KL}(q(x_T|x_0)||p(x_T))$. This term is typically not optimized. Why?

- It is always zero when the network is well-trained
- $q(x_T|x_0)$ is fixed by the noise schedule and equals $N(0,I)$ by design, so $D_{KL}=0$ regardless of θ
- Optimizing it would require the gradient of the Gaussian CDF
- The prior matching term is only relevant during sampling, not training

12. The ELBO for a VAE can be written as $p(x) \geq L$. Why is maximizing the ELBO instead of $p(x)$ directly necessary?

- $p(x)$ is not differentiable with respect to the decoder parameters
- Computing $p(x)=\int p(x|z)p(z)dz$ requires integrating over all z — intractable for continuous latent variables; the ELBO is a tractable surrogate
- $p(x)$ is always negative, making optimization numerically unstable
- The ELBO is always larger than $p(x)$, so maximizing it is more conservative

13. VDM (Video Diffusion Model) ELBO contains a consistency term:

- Temporal motion consistency penalty
- $E[D_{KL}(q(x_{t|x_{t-1}},x_0)||p_{\theta}(x_{t|x_{t+1}}))]$ — ensures learned reverse is consistent with forward process at intermediate t
- Reconstruction loss on keyframes only
- Temporal smoothing regularizer on latent trajectory

14. The ELBO for VDMs (from slides) contains consistency terms $\sum_t D_{KL}(q(x_{t|x_{t-1}},x_0)||p_{\theta}(x_{t|x_{t+1}}))$. These terms encourage:

- The forward process to be faster than the reverse process
- The learned reverse denoising transitions $p_{\theta}(x_{t|x_{t+1}})$ to match the forward process transitions $q(x_{t|x_{t-1}},x_0)$ at every intermediate timestep
- The network to predict the exact noise rather than an approximate score
- The latent x_T to be as far from the data as possible

15. Both VAE and VDM learn a generative model $p_{\theta}(x)$. A key difference in how they do so:

ANSWER KEY

- A. VAE generates by sampling from $N(0, I)$ once; VDM generates by iteratively denoising over T steps from $N(0, I)$
- B. VAE requires labeled data; VDM works unsupervised
- C. VDM uses a discrete latent space; VAE uses a continuous latent space
- D. VAE optimizes the exact log-likelihood; VDM optimizes an upper bound
16. Signal-to-noise ratio $SNR(t) = \frac{\alpha_t}{1 - \alpha_t}$ as $t \rightarrow T$:
- A. $SNR \rightarrow \infty$ (pure signal)
- B. $SNR \rightarrow 0$ — $\{\alpha_t\}_{t \rightarrow 0}$, signal fully destroyed
- C. $SNR \rightarrow 1$
- D. SNR oscillates
17. VAE decoder $p_\theta(x|z) = N(x; \mu_\theta(z), \sigma^2 I)$ (fixed σ): reconstruction loss:
- A. Cross-entropy
- B. MSE: $\|x - \mu_\theta(z)\|^2 / (2\sigma^2)$ — from $-p(x|z)$ for Gaussian likelihood
- C. L_1 loss
- D. Perceptual (VGG) loss
18. The ELBO $L = E_{q(z|x)}[-D_{KL}(q(z|x)||p(z))]$ is a lower bound on $p(x)$. This is because:
- A. $D_{KL}(q||p) \leq 0$ always, so the ELBO is larger than $p(x)$
- B. $D_{KL}(q(z|x)||p(z|x)) \geq 0$ always, and $p(x) = L + D_{KL}(q(z|x)||p(z|x))$
- C. The reconstruction term is always less than zero, bringing the ELBO below $p(x)$
- D. The expectation over $q(z|x)$ introduces a bias that makes $L > p(x)$
19. In the DDPM reverse process, the network $\epsilon_\theta(x_t, t)$ predicts the noise to enable denoising. The mean of the reverse step $p_\theta(x_{t-1}|x_t)$ is:
- A. x_t — the current noisy image unchanged
- B. $\sqrt{1 - \alpha_t}(x_t - \beta_t) + \sqrt{1 - \alpha_t}\epsilon_\theta(x_t, t)$
- C. $\{\alpha_t\}_{t=1}x_0 + \{1 - \alpha_t\}_{t=1}\epsilon$
- D. $x_t - \epsilon_\theta(x_t, t)$

- T1 Basic Image Operations:** 1:B 2:C 3:C 4:B 5:C 6:C 7:B 8:B 9:C 10:B 11:B 12:B 13:B 14:C 15:B 16:B 17:B 18:B 19:B 20:B
- T2 Edge Detection:** 1:B 2:B 3:B 4:A 5:B 6:B 7:B 8:B 9:A 10:B 11:A 12:B 13:B 14:B 15:A 16:B 17:B 18:B 19:B 20:B
- T3 Fitting: Hough & RANSAC:** 1:B 2:B 3:B 4:B 5:C 6:B 7:B 8:B 9:B 10:B 11:A 12:B 13:B 14:B 15:B 16:B 17:B 18:B 19:B 20:B
- T4 Sampling, Aliasing & Pyramids:** 1:B 2:B 3:B 4:B 5:B 6:B 7:B 8:B 9:C 10:B 11:C 12:A 13:B 14:B 15:B 16:B 17:B 18:B 19:B 20:B
- T5 Deep Learning Fundamentals:** 1:B 2:B 3:B 4:B 5:B 6:B 7:B 8:B 9:C 10:B 11:B 12:B 13:B 14:B 15:B 16:B 17:B 18:B 19:B 20:B
- T6 CNN Architectures:** 1:B 2:B 3:B 4:B 5:B 6:B 7:B 8:B 9:B 10:B 11:B 12:B 13:B 14:B 15:B 16:A 17:B 18:B 19:B 20:B
- T7 Image Segmentation:** 1:B 2:B 3:B 4:B 5:B 6:B 7:B 8:B 9:B 10:B 11:B 12:B 13:B 14:B 15:B 16:B 17:B 18:B 19:B 20:B
- T8 Motion Estimation & Optical Flow:** 1:B 2:B 3:A 4:B 5:B 6:B 7:B 8:B 9:B 10:B 11:B 12:B 13:A 14:A 15:C 16:B 17:B 18:B 19:B
- T9 Tracking & Particle Filters:** 1:B 2:B 3:A 4:B 5:B 6:B 7:B 8:B 9:B 10:C 11:B 12:B 13:B 14:B 15:B 16:B 17:C 18:B 19:B 20:B
- T10 Camera Models & Projective Geometry:** 1:B 2:B 3:B 4:B 5:B 6:B 7:B 8:B 9:B 10:B 11:A 12:B 13:B 14:B 15:B 16:B 17:B 18:B 19:A 20:C
- T11 Stereo Vision & Structure from Motion:** 1:B 2:B 3:A 4:B 5:C 6:B 7:C 8:C 9:B 10:B 11:B 12:B 13:B 14:B 15:B 16:B 17:B 18:B 19:B 20:B
- T12 Generative Models:** 1:B 2:B 3:B 4:B 5:A 6:B 7:B 8:B 9:B 10:B 11:B 12:B 13:B 14:B 15:A 16:B 17:B 18:B 19:B